PREDICTING THE CONSTRUCTION COST OF NUCLEAR REACTORS IN FRANCE

Introduction

- **Background**: France's large-scale nuclear expansion programme during the period.
- **Dataset**: 58 French nuclear reactors in 1970s–1980s.
- **Aim**: Predict construction cost of future reactors.

Methodology

- Multiple Linear Regression (**MLR**)
- Backward stepwise (**AIC**/**BIC**)

Variable (unit)	Interpretation
Location (No.1–19)	Coded site ID where the reactor was built
Cost (million francs)	Construction cost of the reactor
Gap_days (days)	Days elapsed since the earliest project start
App_time (months)	Length of application process
Build_time (months)	Length of construction process
Capacity (Mwe)	Power plant net capacity
JP (0/1)	Joint planning with another reactor
N (units)	Number of power plants constructed by architect

Steps

01 Data Cleaning & Transformation

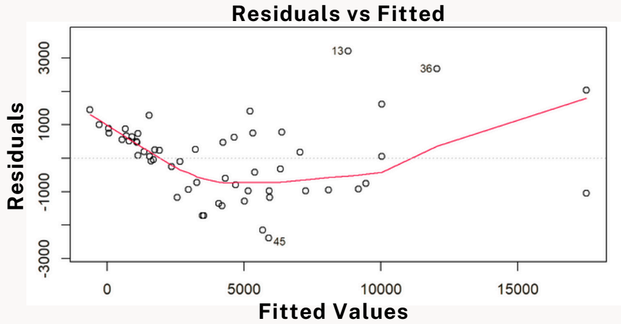
- Excluded Civaux data (outlier).
- Removed *Location*
- Converted *date* into *gap_days*

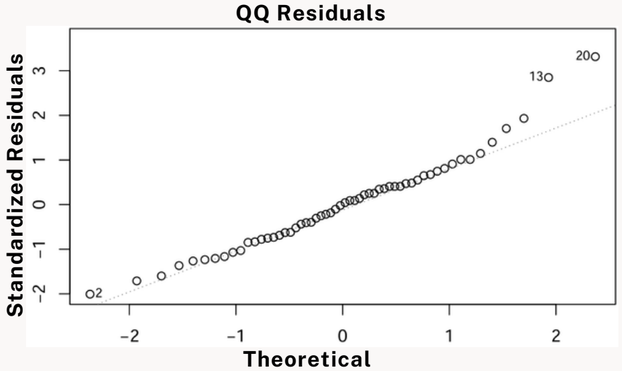
02 Initial Model

- Backward stepwise regression
- Predictors: app_time, build_time, capacity, N, JP, gap_days
- Model fit: R^2=0.922, Adjusted R2=0.913

03 Model Diagnostics

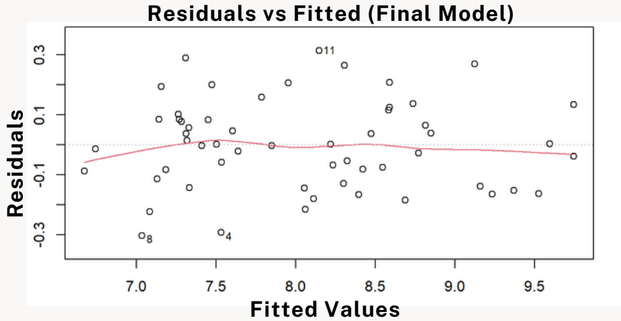
- Residuals plots, Q–Q Plot, Scale–Location





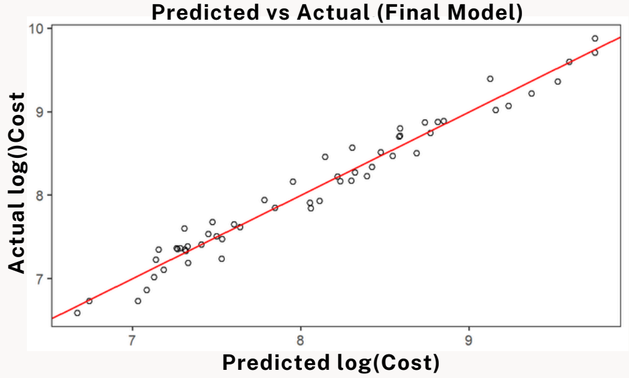
04 Model Adjustment

- Checked scatterplots for each predictor
- Cost → log(Cost) (stabilize variance)
- Dropped app_time (weak explanatory power) and capacity (insignificant after log-transformation)



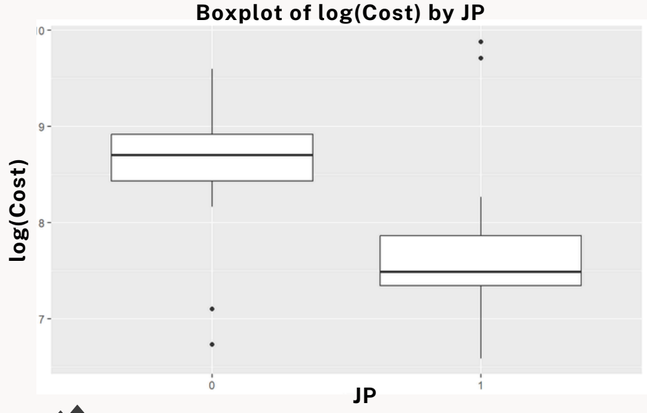
05 Final Model

- Log(Cost) = 6.58 + 0.0058 · build_time − 0.015 · N − 0.29 · JP + 0.00052 · gap_days
- All predictors statistically significant ($p < 0.001$)
- Model fit: R2=0.967, Adjusted R2=0.965



Interpretation

- Longer **construction time** and **later project starts** raise costs.
- More **experience (N)** reduces costs.
- **Joint planning** yields the largest savings.



limitations

- Data from 56 French reactors only → **limited generalisability**.
- **Economic and policy factors** not included.
- Residuals show slight **non-linearity**.
- **Future costs** may **differ** from historical patterns.

Conclusions

- Log-linear model fits well (Adj. $R^2 \approx 0.97$).
- Main drivers: **build time**, **project timing**, **experience (N)**, **joint planning (JP)**.
- **Longer builds** and **later starts raise** costs; **more experience lowers** them.
- **Joint planning (JP)** gives the largest savings.

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